

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 1 SET A

SUMMER SEMESTER, 2023-2024

DURATION: 25 MINUTES

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

1. A number $(891)_b$ in an unknown base b is equivalent to the hexadecimal number $(4ED)_{16}$. Determine the base b .

3
(CO1)
(PO1)

Solution:

Given:

$$(891)_b = (4ED)_{16}$$

Step 1: Convert hexadecimal to decimal

$$(4ED)_{16} = 4 \cdot 16^2 + 14 \cdot 16^1 + 13 \cdot 16^0 = 1024 + 224 + 13 = 1261$$

Step 2: Convert $(891)_b$ to decimal and solve

$$8b^2 + 9b + 1 = 1261 \Rightarrow 8b^2 + 9b - 1260 = 0$$

Using the quadratic formula:

$$b = \frac{-9 \pm \sqrt{9^2 - 4 \cdot 8 \cdot (-1260)}}{2 \cdot 8} = \frac{-9 \pm \sqrt{81 + 40320}}{16} = \frac{-9 \pm \sqrt{40401}}{16} = \frac{-9 \pm 201}{16}$$

Evaluating both roots:

$$b = \frac{-9 + 201}{16} = \frac{192}{16} = \boxed{12}$$

$$b = \frac{-9 - 201}{16} = \frac{-210}{16} = -13.125 \quad (\text{Not valid})$$

Final Answer:

$$\boxed{b = 12}$$

Rubrics:

- 1 point for correctly converting the two numbers to a known base.
- 1 point for correctly expressing the given numbers into a valid equation.
- 1 point for correctly determining the root of the equation and neglecting the invalid root.

2. Convert $(38)_{10}$ into Gray Code.

3
(CO1)
(PO1)

Solution:

At first convert the given number into binary number,

$$(38)_{10} = (100110)_2$$

Final Answer:

$$(100110)_2 = (110101)_{Gray}$$

Rubrics:

- 1 point for correctly converting the decimal number into binary number.
- 1 point for writing down relevant equation/expression to convert binary number to gray code.
- 1 point for correct gray code.

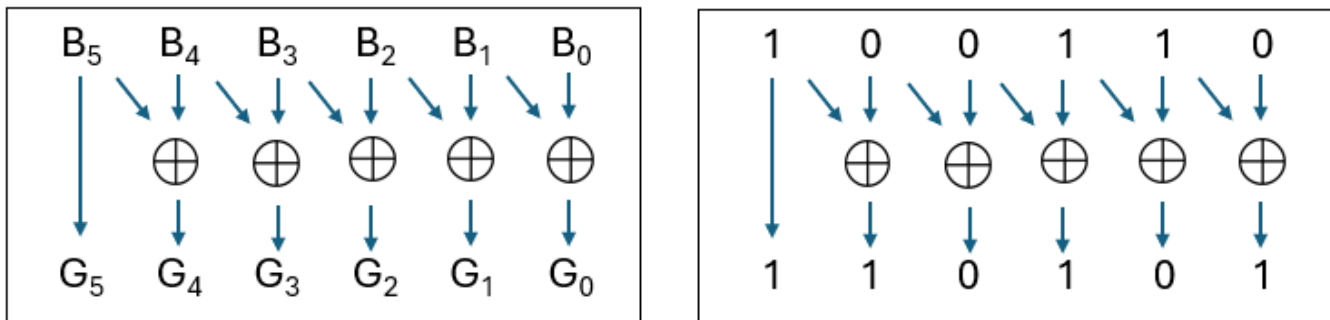


Figure 1: Binary to Gray

3. Subtract $(645)_9$ from $(847)_9$ using *Diminished Radix Complement*. (Use a 4 digit register.)

6
(CO2)
(PO1)

Solution:

Given,

$$(847)_9 - (645)_9$$

Convert to 4 digit register compatible

$$= (0847)_9 + (-0645)_9$$

Converting $(0645)_9$ to (9-1) or 8's Complement (subtract each digit from 8),

$$8\text{'s Complement of } (0645)_9 = (8243)_{8\text{'s Complement}}$$

So,

$$(847)_9 - (645)_9 = (202)_9$$

Final Answer:

$$(202)_9$$

Rubrics:

- 1 point for using 4 digit representation.
- 1 point for representing the subtraction operation in terms of addition.
- 1 point for correctly performing 8's complement.
- 2 points for correct addition using 8's complement.
- 1 point for correctly handling the overflow bit.

$(0847)_9$	→	$\begin{array}{cccc} & 1 & 1 & 1 \\ 0 & 8 & 4 & 7 \end{array}$
$(0645)_9$'s 8's complement	→	$\begin{array}{cccc} 8 & 2 & 4 & 3 \end{array}$
	(+)	
		$\begin{array}{cccc} 1 & 0 & 2 & 0 & 1 \\ \text{(overflow)} & & & & \\ \text{DISCARD} & & & & \\ \text{Add 1} & & & & \end{array}$
		$\begin{array}{cccc} & & & + & 1 \\ \hline & & & 2 & 0 & 2 \end{array}$

$\therefore (847)_9 - (645)_9 = (202)_9$

Figure 2: 8's Complement

4. Simplify the following Boolean expression using a Karnaugh Map (K-map):

$$F(A, B, C, D) = \prod M(3, 6, 9, 12, 15) + \sum d(0, 1, 2)$$

Where:

- $\prod M(\dots)$ denotes a product of maxterms.
- $\sum d(\dots)$ denotes the don't-care conditions.

While solving, draw the K-Map and identify the groups(minterms).

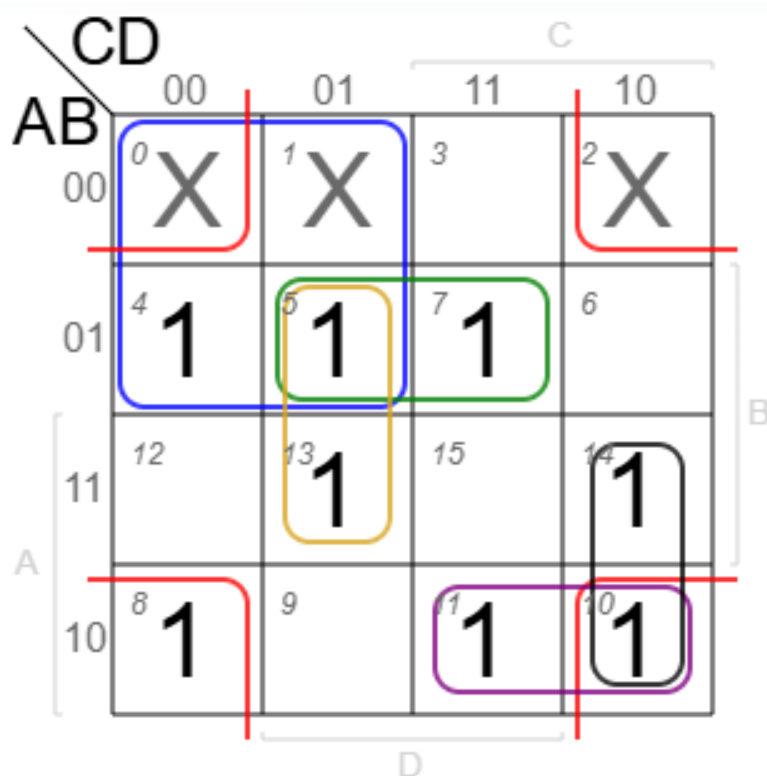
8
(CO3)
(PO2)

Solution:**Final Answer:**

$$F(A, B, C, D) = A'C' + B'D' + A'BD + BC'D + AB'C + ACD'$$

Rubrics:

- 2 point for building the 4 variable K-map.
- 1 point for correctly identifying each essential prime component (times 6).



$$F = A'C' + B'D' + A'BD + BC'D + AB'C + ACD'$$

Figure 3: K-map